Decision Transformer for Limit Order Book Trading

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Reinforcement Learning Framework

RL formalises sequential decision-making as an agent interacting with an environment over discrete timesteps. At each step t, the agent observes state s_t, selects action $a_t \sim \pi(\cdot \mid s_t)$, and receives reward r_t. The goal is to find a policy π^* that maximises the expected cumulative discounted return:

$$J(\pi) = \mathbb{E}_\pi\Big[\sum_{t=0}^{T} \gamma^t r_t\Big], \quad \gamma \in [0,1)$$

From online to offline RL. Classical RL (Q-learning, PPO, SAC) requires live interaction with the environment. This is costly or impossible in finance. **Offline RL** (also called batch RL) learns entirely from a fixed dataset $\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}$ collected by *behaviour policies*, without any further environment interaction.

Key challenge of offline RL: *distributional shift*, i.e. the learned policy may query (s, a) pairs unseen in $\mathcal{D}$, leading to overestimated Q-values. The **Decision Transformer** sidesteps this by recasting the problem as supervised sequence modelling, avoiding value function estimation entirely.

Return-to-go conditioning. Instead of learning $V(s)$ or $Q(s,a)$, the DT is conditioned on the desired future return $\hat{R}_t = \sum_{t'=t}^{T} r_{t'}$ specified at inference time. This allows direct *behavioral control* without re-training:

$$a_t = \pi_\theta\big(a_t \mid \hat{R}_t,\ s_t,\ a_{t-1},\ \hat{R}_{t-1},\ s_{t-1},\ \dots\big)$$

Motivation & Problem Statement

Financial markets match buyers and sellers via a **Limit Order Book (LOB)**, an electronic table of bid and ask orders at multiple price levels.

Why offline RL for LOB trading?

- Live trading experiments are expensive and risky
- Historical LOB data is abundant and rich
- Reward signal is sparse: profitable moves are rare
- Classical RL diverges on non-stationary financial data

Our approach: train a Decision Transformer on $\sim$1.2M offline transitions from 6 heuristic policies, then condition it at inference on varying target returns to control aggressiveness.

Related Works

FI-2010 dataset (Ntakaris et al., arXiv:1705.03233) – benchmark LOB data for 5 Finnish stocks, Z-score normalised, 10 price levels, horizon labels $k \in \{1, 2, 3, 5, 10\}$.

DeepLOB (Zhang et al., 2019) – CNN+LSTM architecture reaching state-of-the-art mid-price prediction on FI-2010; defines our feature representation $\mathbf{x}_t \in \mathbb{R}^{40}$.

Decision Transformer (Chen et al., arXiv:2106.01345) – reformulates RL as sequence modelling; a GPT-2 style causal transformer predicts the next action conditioned on $(\hat{R}_t, s_t, a_t)$ triplets.

Trading Environment (MDP)

We implement a Gymnasium environment `LOBTradingEnv`:

State s_t	LOB snapshot $\mathbf{x}_t \in \mathbb{R}^{40}$ + position $p_t \in \{-1, 0, 1\}$
Action a_t	{Short, Flat, Long}
Reward r_t	Mid-price return $\times$ position, minus transaction cost
Horizon	2 000 steps (training) / full day (test)

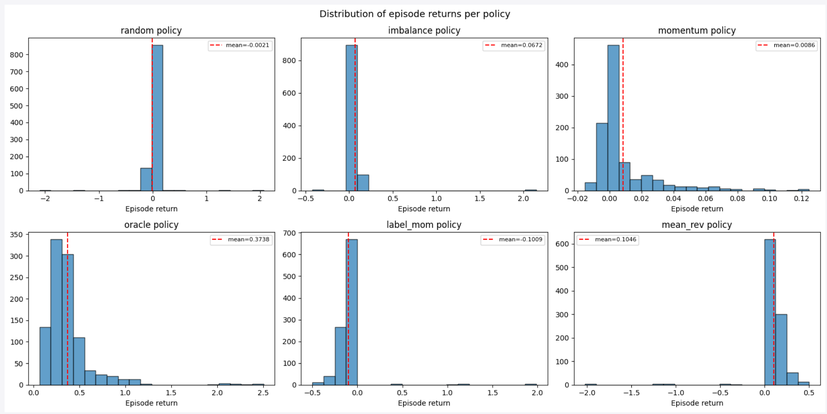
The state dimension fed to the Decision Transformer is $d_s = 41$ (40 LOB features + current position).

Offline Dataset Collection

We generate $\sim$6000 trajectories ($\approx$ 1.2M transitions) from **6 heuristic policies** on random training windows:

- `random` – uniform random action
- `imbalance` – trades on bid/ask volume imbalance
- `momentum` – follows short-term price trend
- `mean_reversion` – fades recent moves
- `oracle` – uses future labels, `hold` – stays flat

After each episode, return-to-go $\hat{R}_t = \sum_{t'=t}^{T} r_{t'}$ is computed and stored alongside (s_t, a_t, r_t).



Return distributions across 6 offline policies. Oracle mean = 0.37, mean_rev = 0.10, imbalance = 0.07; random and label_mom cluster near zero or negative.

Decision Transformer Architecture

The DT takes a context window of K timesteps as input. Each timestep contributes **three tokens**: return-to-go $\hat{R}_t$, state s_t, and action a_t.

Causal Self-Attention. Input $\mathbf{X} \in \mathbb{R}^{T \times d}$ is projected into queries, keys, values:

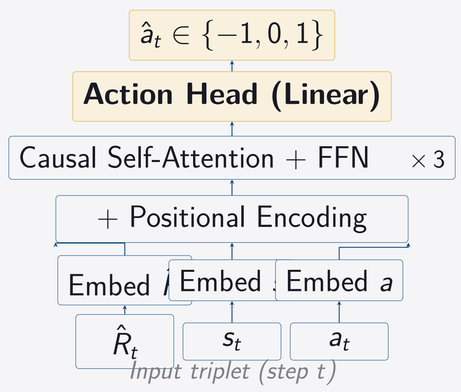
$$\mathbf{Q} = \mathbf{X}W_Q,\ \mathbf{K} = \mathbf{X}W_K,\ \mathbf{V} = \mathbf{X}W_V$$

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}$$

where $\mathbf{M}$ is a causal (upper-triangular) mask.

Transformer Block (GPT-2 pre-norm):

$$\mathbf{h} = \mathbf{x} + \mathrm{Attn}(\mathrm{LN}(\mathbf{x})), \quad \mathbf{y} = \mathbf{h} + \mathrm{MLP}(\mathrm{LN}(\mathbf{h}))$$



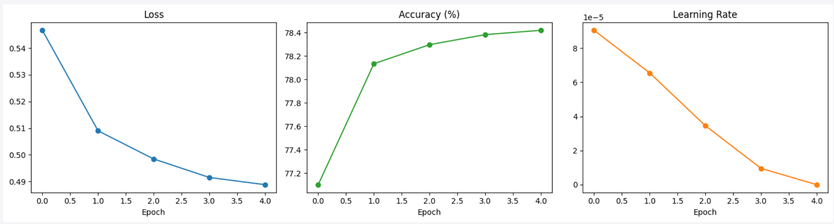
Hyperparameter	Value
Context length K	20 timesteps
Embedding dim d	128
Transformer blocks N	3
Attention heads	4
Dropout	0.1

Training

Objective. Cross-entropy loss on the predicted action token, computed only on state-position tokens (action tokens at each step):

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|} \sum_{(s,a)\in\mathcal{B}} \log \hat{\pi}_\theta(a_t \mid \hat{R}_t, s_t)$$

Setup: 5 epochs, batch size 1024, Adam ($\eta = 10^{-4}$, $\lambda = 10^{-4}$), cosine LR decay, gradient clipping at 1.0. Final loss: 0.489, final accuracy: **78.4**% on offline action labels.



Loss (0.55 → 0.49) and accuracy (77.1% → 78.4%) over 5 epochs. LR decays with cosine schedule.

Evaluation Protocol

The model is evaluated on **day 10** of FI-2010 (held-out, never seen during training) in autoregressive rollout mode.

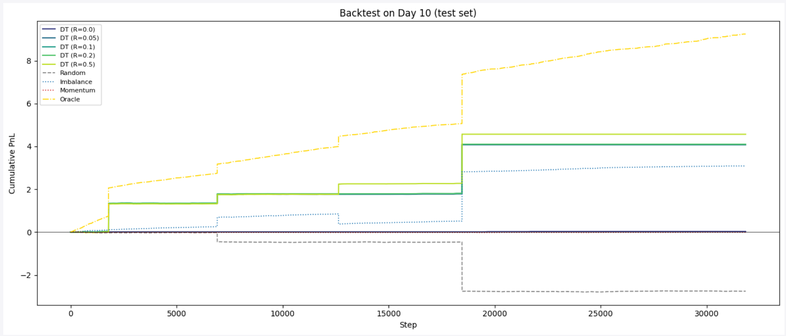
Target-return conditioning. At inference we condition the DT on a fixed target return $\hat{R}_0 \in \{0.05, 0.10, 0.20, 0.30\}$ and observe how the trading behavior changes.

Baselines: random, imbalance, momentum, mean-reversion, hold, and oracle (using true future labels as upper bound).

Key Results

The Decision Transformer **outperforms all non-oracle heuristics** on the held-out test day across all conditioning values $\hat{R} \geq 0.05$. The best variant ($\hat{R} = 0.5$) achieves **PnL** $+4.57$**, Sharpe** $+2.94$, with a max drawdown two orders of magnitude below Imbalance (0.042 vs 0.477). Conditioning on $\hat{R} = 0$ collapses performance to near-zero PnL, confirming that return-to-go is an effective behavioral control signal.

Cumulative PnL on Test Set

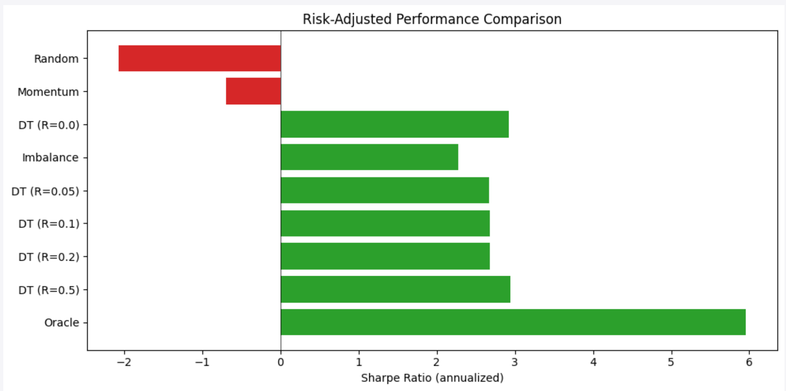


Cumulative PnL over the full test day. DT variants (blue gradient) vs heuristic baselines. Oracle shown as upper bound (dashed).

Performance Metrics

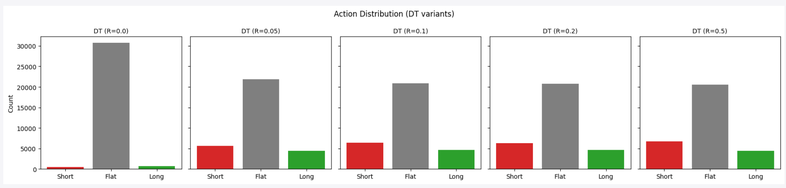
Agent	PnL	Sharpe	Max DD
Oracle	$+9.24$	$+5.96$	0.000
DT ($\hat{R} = 0.50$)	$+4.57$	$+2.94$	0.042
DT ($\hat{R} = 0.20$)	$+4.10$	$+2.68$	0.024
DT ($\hat{R} = 0.10$)	$+4.09$	$+2.68$	0.021
DT ($\hat{R} = 0.05$)	$+4.07$	$+2.67$	0.020
DT ($\hat{R} = 0.00$)	$+0.03$	$+2.92$	0.012
Imbalance	$+3.09$	$+2.27$	0.477
Momentum	-0.01	-0.70	0.039
Random	-2.76	-2.07	2.795

Sharpe Ratio Comparison



Sharpe ratio per agent. Positive = green, negative = red. DT with optimal conditioning achieves the highest risk-adjusted return among trainable agents.

Action Distribution



Action frequencies (Short / Flat / Long) for each DT target return. Higher conditioning shifts the agent toward more active positioning.

Summary & Takeaways

Offline RL via Decision Transformer is a viable framework for LOB trading without any online environment interaction.

Return-to-go conditioning acts as a **behavioral dial**: increasing the target return leads to more aggressive positioning, up to a ceiling set by market predictability.

All DT variants with $\hat{R} > 0$ surpass heuristic baselines on both PnL and Sharpe. The best DT ($\hat{R} = 0.5$) reaches 49% of oracle PnL with Sharpe $+2.94$, vs $+2.27$ for Imbalance, and a max drawdown $10\times$ lower (0.042 vs 0.477).

Limitations: no transaction costs modelled during training; FI-2010 is a lab dataset (5 stocks, 10 days); no fine-grained hyperparameter search.

Future work: online fine-tuning, multi-asset generalisation, incorporating order-book depth dynamics as additional features.